

EE251 Lab 3

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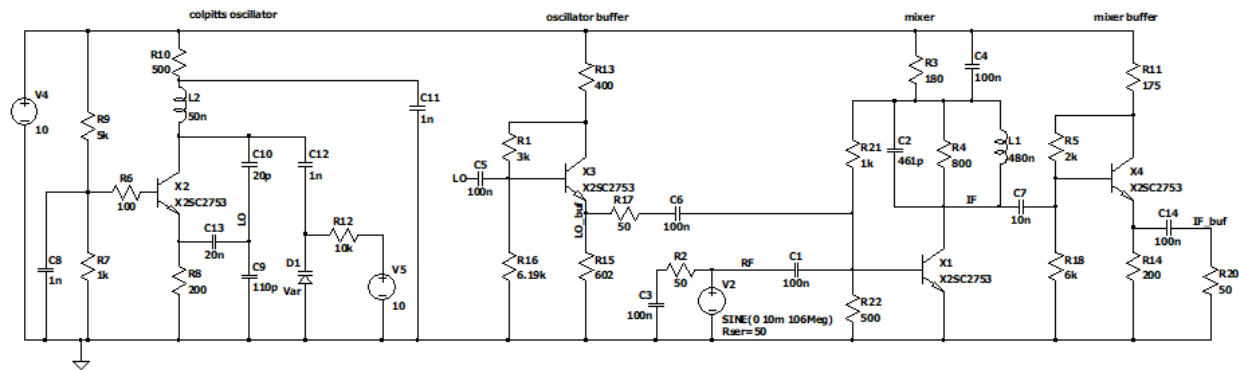
Date: 02.04.2026

Oscillator and Mixer Result

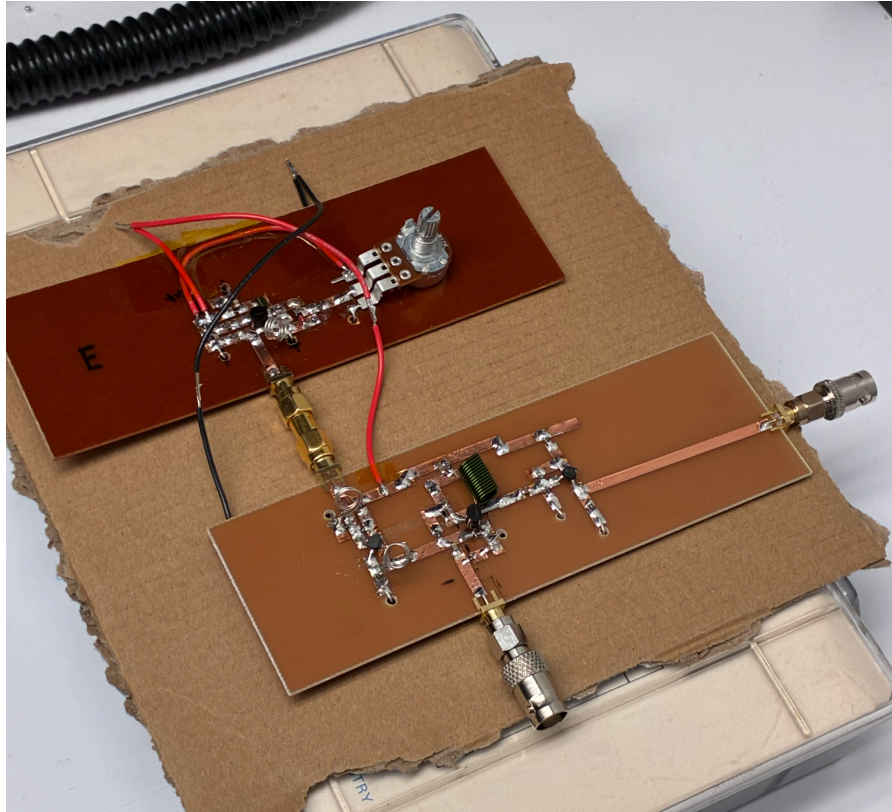
Conversion Gain (-23 dBm RF input)	Mixer output bandwidth
-13 dBm at 10.7 MHz	~250 KHz

	10.7 MHz	88 MHz	98 MHz	108 Mhz
S11 (dB)	NA	-15.2	-18.8	-16.3
S22 (dB)	-13.4	NA	NA	NA

Oscillator and Mixer Schematic



Physical Board



Technical Discussion

Local Oscillator (LO):

For the LO we opted for a Colpitts design using a varactor for frequency tuning. The desired tuning range is from 98.7 to 118.7 MHz. To achieve this range we first chose our inductor to be 50nH. The resonant frequency of the Colpitts is computed with the following equation.

$$\omega_0 = \frac{1}{\sqrt{LC_{eq}}}$$

C_{eq} is the equivalent capacitance of the oscillator. Typically the oscillator has a capacitor divider network so $C_{eq} = C_1 \parallel C_2$. However since we need to make the oscillator tunable we introduce a varactor in parallel to the capacitor divider network. Thus the equivalent capacitance is $C_{eq} = C_1 \parallel C_2 + C_{var}$. In the schematic the varactor is in series with a 1nF capacitor. This is to block the bias voltage from the oscillator. The blocking capacitor is magnitudes larger than the varactor so their series value will be dominated by the varactor since it is smaller.

We used a 1N5248 zener diode to act as our variable capacitor. It ranges from 18pF to 36pF with a 1V to 10V bias swing. With this bound we iteratively computed C_1 . First we let C_2 be

110pF. We are shooting for a C_2/C_1 ratio of about 4 and we found that this value for C_2 works reasonably well. Then we varied C_1 such that the upper and lower bound roughly matched the desired tuning range. We found that $C_1 = 20pF$ gives us the desired range.

Computing the lower frequency bound is the following.

$$C_{eq} = \frac{1}{\omega_0^2 L} = \frac{1}{(2\pi f_0)^2 L} = \frac{1}{(2\pi \cdot 98.7MHz)^2 \cdot 50nH} = 52.7pF$$

$$C_{var} = C_{eq} - C_1 \parallel C_2 = 52.7pF - 20pF \parallel 110pF = 35.8pF$$

Since C_{var} is less than the maximum value for the varactor, the current values for C_1 and C_2 satisfy the lower frequency bound. Computing the upper frequency bound is the following.

$$C_{eq} = \frac{1}{(2\pi \cdot 118.7MHz)^2 \cdot 50nH} = 35.2pF$$

$$C_{var} = C_{eq} - C_1 \parallel C_2 = 35.2pF - 20pF \parallel 110pF = 18.3pF$$

Since C_{var} is more than the minimum value for the varactor, the current values for C_1 and C_2 satisfy the upper frequency bound.

Mixer:

Initially we chose a mixer configuration that fed the RF to the base and the LO to the emitter. Our main motivation was that this would provide better isolation for the oscillator from the RF injection. However, we were not able to get enough conversion gain, so we opted to connect both the RF and LO at the base. This helped to increase our conversion gain because of the ease in matching at the base as opposed to the emitter. Ultimately more of the LO signal is passed through which creates stronger mixing products.

The mixer is essentially a narrow band amplifier. The resonant frequency of the tank needs to be 10.7MHz. For high conversion gain and high Q we iteratively found that $L = 480nH$. Then using the resonant frequency we can calculate the tank's capacitor.

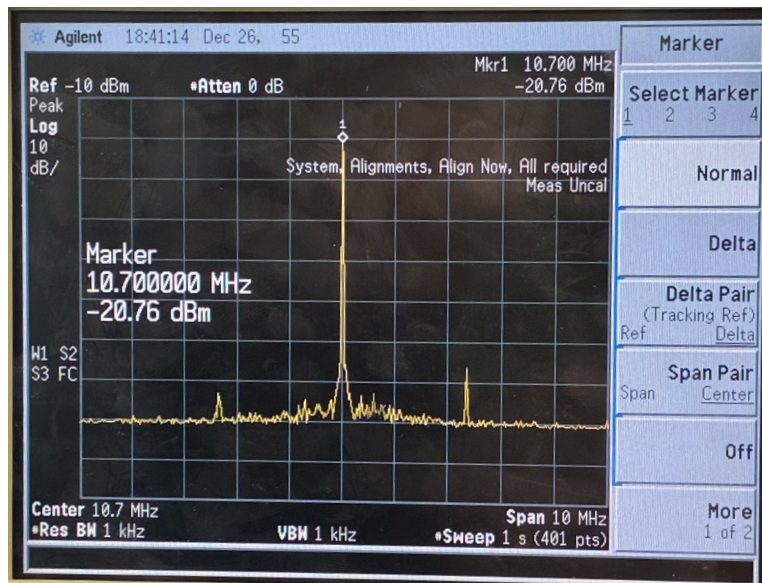
$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(2\pi \cdot 10.7MHz)^2 \cdot 480nH} \approx 460pF$$

For high Q we iteratively found that a tank resistance of 800Ω worked well. Above the tank is a current limiting resistor with an AC bypass.

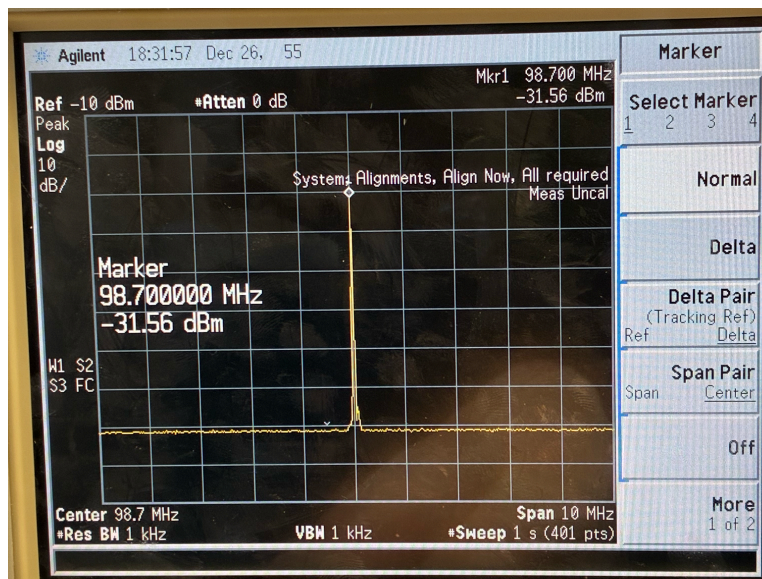
LO and Mixer Buffer:

Between the LO and the mixer and the mixer to the output is a buffer. We use buffers to provide extra isolation between the buffer and the mixer and the mixer to the output. They are also convenient for impedance matching. The buffers are biased to present the input with a high impedance so that it will load it and change its behavior. In addition it also has a low output impedance which is useful for 50Ω output impedance matching for the next stage.

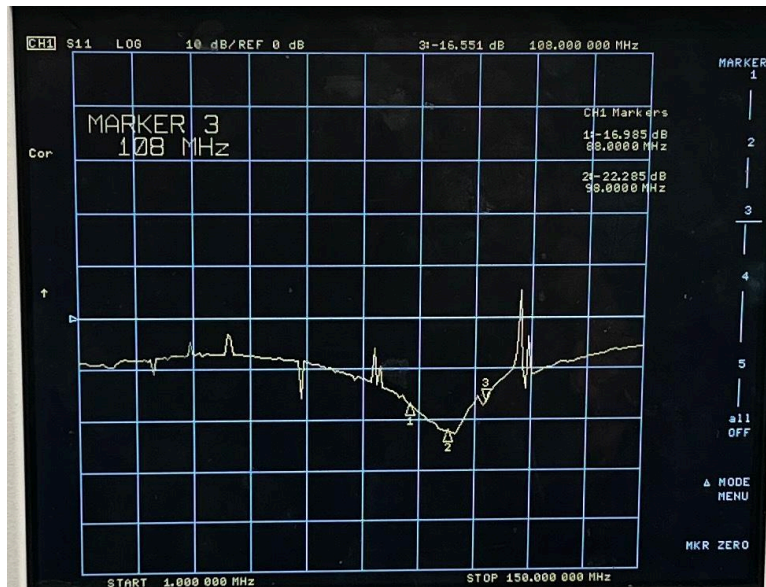
Mixer Output Spectrum Analyzer Plot (RF Input: -31.56dbm)



RF Input Spectrum Analyzer Plot



Mixer RF Input VNA S11 Plot



Mixer RF Output VNA S11 Plot

